

Business Understanding Report: Customer Churn Analysis

Business Context

Customer churn occurs when customers discontinue their services. In telecommunications, reducing churn is important because customer loss can reduce recurring revenue and customer lifetime value. Industry research highlights the use of customer analytics to identify churn patterns and improve retention.

Business Problem

The business needs to understand which customer characteristics are associated with churn so that retention efforts can be targeted toward higher-risk segments instead of treating all customers equally.

Objective

This project analyzes 7,043 telecom customer records to identify churn patterns and provide actionable recommendations for improving customer retention.

The analysis focuses on:

- Customer tenure
- Contract type
- Monthly charges
- Internet service
- Payment method
- Customer demographics

Key Business Questions

1. What is the overall churn rate?
2. Which customer segments have the highest churn?
3. How are tenure and contract type associated with churn?
4. Are higher monthly charges associated with greater churn?

5. Which internet services and payment methods show higher churn?
6. Which combinations of characteristics represent the highest-risk customers?

Key KPI

Overall Churn Rate: 26.54%

- Total customers: **7,043**
- Churned: **1,869**
- Retained: **5,174**

Stakeholders

The findings can support customer retention, marketing, customer service, product, and management teams by identifying high-risk segments and potential areas for intervention.

Expected Business Value

The analysis can help the business develop targeted retention strategies. Initial findings show that customers with 0–12 months of tenure on month-to-month contracts have a 51.35% churn rate, while fiber-optic and electronic-check customers also show elevated churn.

These findings indicate where further investigation and targeted retention efforts should be focused. The analysis identifies associations rather than causation.

Analytical Approach

Python was used to inspect, clean and analyze the dataset and create visualizations including bar charts, pie charts, histograms, a box plot and correlation heatmap. The findings were then translated into business insights, risks, opportunities and recommendations.